

Down Payments, Deferred Homes: How LTV Restrictions Reshape Household Consumption

Yann Cerasi
Zurich & SFI

Gazi Kabas
Tilburg

Steven Ongena
Zurich & SFI

Kasper Roszbach
Norges Bank & Lund

The views expressed here are those of the authors, and not necessarily those of the Norges Bank.

Housing markets and consumption are tightly linked

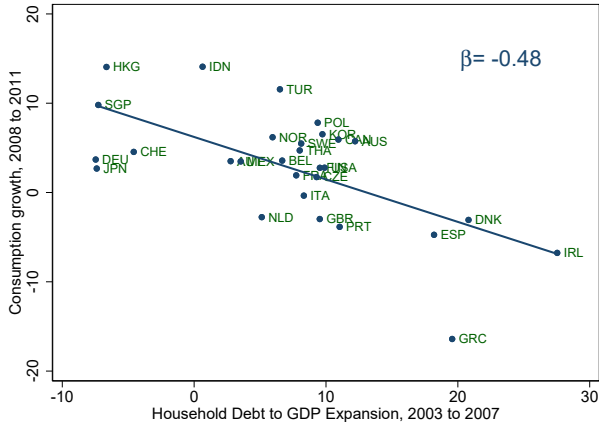
- House prices affect consumption via wealth and collateral channels
(Campbell & Cocco 2007; Mian, Rao & Sufi 2013; Guren et al. 2021)
- This link is one of the main drivers of business-cycle fluctuations and credit-driven booms
(Bernanke & Gertler 1986; Kiyotaki & Moore 1997; Iacoviello 2005)

Housing markets and consumption are tightly linked

- House prices affect consumption via wealth and collateral channels
(Campbell & Cocco 2007; Mian, Rao & Sufi 2013; Guren et al. 2021)
- This link is one of the main drivers of business-cycle fluctuations and credit-driven booms
(Bernanke & Gertler 1986; Kiyotaki & Moore 1997; Iacoviello 2005)

Via this link, household leverage amplifies the recessions

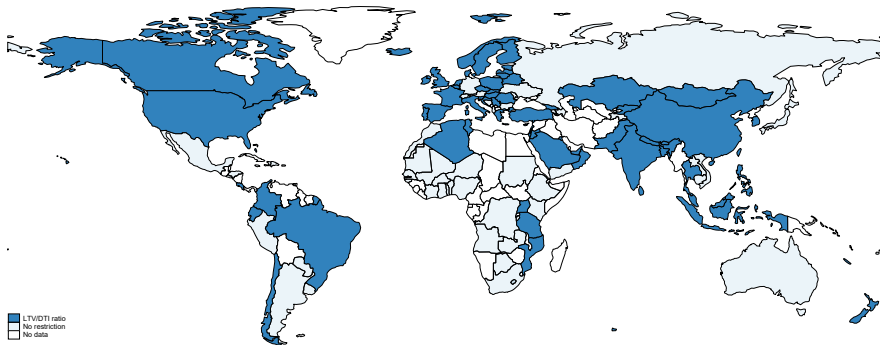
Housing and consumption



Increase in HH debt before the GFC predicts a decline in consumption during GFC



Macroprudential Policies in 2021



Many countries have implemented borrowing restrictions on HH

Theoretical mechanism

How do prospective homebuyers (renters) adjust consumption in response to tighter borrowing constraints?

Important to understand the aggregate demand implications of these constraints

How do prospective homebuyers (renters) adjust consumption in response to tighter borrowing constraints?

Important to understand the aggregate demand implications of these constraints

Standard life-cycle models with binding borrowing constraints

- Tighter LTV $\Rightarrow$ higher down payment
- Renters *increase* savings to meet requirement
- Consumption *falls*

Artle & Varaiya 1978; Slemrod 1982; Jappelli & Pagano 1994

How do prospective homebuyers (renters) adjust consumption in response to tighter borrowing constraints?

Important to understand the aggregate demand implications of these constraints

Standard life-cycle models with binding borrowing constraints

- Tighter LTV $\Rightarrow$ higher down payment
- Renters *increase* savings to meet requirement
- Consumption *falls*

Artle & Varaiya 1978; Slemrod 1982; Jappelli & Pagano 1994

Models with consumption smoothing and endogenous tenure choice

- Tighter LTV $\Rightarrow$ larger required savings
- Renters *delay* their purchase and saving because they expect higher income in the future
- Consumption *rises*

Balke et al. 2024

$\Rightarrow$ **Fundamentally different implications for aggregate demand**

- **Renters increase their consumption after the introduction of LTV restriction**
→ 2.8 pp increase in the consumption/income ratio

- **Renters increase their consumption after the introduction of LTV restriction**
→ 2.8 pp increase in the consumption/income ratio
- **Mechanism:** Delayed home purchase with delayed savings
→ Larger effects when earnings potential is high controlling for current income

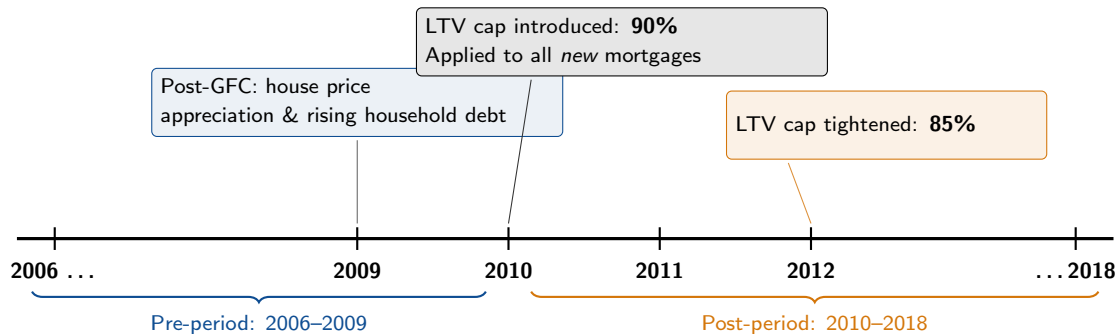
- **Renters increase their consumption after the introduction of LTV restriction**
 - 2.8 pp increase in the consumption/income ratio
- **Mechanism:** Delayed home purchase with delayed savings
 - Larger effects when earnings potential is high controlling for current income
- **Homebuyers** decrease their consumption before and after a home purchase
 - Evidence for rebuilding liquidity buffers

Data & Empirical Strategy

- Consumption, housing transactions, tax filings, individual characteristics
→ Consumption: Debit card transactions, digital invoice, direct remittances.
Approximately 80% of the card payments. 22 COICOP categories
- Sample: 2006-2018, Annual, Household-level

- Consumption, housing transactions, tax filings, individual characteristics
 - Consumption: Debit card transactions, digital invoice, direct remittances. Approximately 80% of the card payments. 22 COICOP categories
- Sample: 2006-2018, Annual, Household-level
- **3 groups:** Renters, homebuyers, homeowners
 - Renters: Do not have housing wealth, no housing transactions before the restriction
 - Homebuyers: First-time homebuyers
 - Homeowners: Have housing wealth before 2006

The Norwegian LTV Restriction



- LTV regulation introduced in **2010**
- Restriction applies only to *new* mortgages $\Rightarrow$ existing homeowners unaffected

$$y_{it} = \beta Renter_i \times Post_t + \gamma_1 Renter_i + \gamma_2 Post_t + \epsilon_{it}$$

- y_{it} = Consumption / Income
- $Renter_i$: =1 if HH is a renter until 2010, 0 if HH is already a homeowner prior to 2006
→ Renters include HHs that purchase a home after the restriction
- $Post_t$: =1 if year $\geq$ 2010

$$y_{it} = \beta Renter_i \times Post_t + \gamma_1 Renter_i + \gamma_2 Post_t + \epsilon_{it}$$

- y_{it} = Consumption / Income
- $Renter_i$: =1 if HH is a renter until 2010, 0 if HH is already a homeowner prior to 2006
→ Renters include HHs that purchase a home after the restriction
- $Post_t$: =1 if year ≥ 2010
- **Identifying assumption**
 1. Without the restriction, the consumption/income difference between renters and homeowners would be the same
→ The levels can be different
 2. Homeowners are not affected by the restriction
→ The restriction can reduce homeowners' access to credit, or housing wealth

Results

Consumption Reaction of Renters

	Consumption/Income		
	(1)	(2)	(3)
Renter $\times$ Post	0.0318*** (0.00349)	0.0296*** (0.00374)	0.0276*** (0.00340)
Renter	-0.0837*** (0.00286)	-0.0814*** (0.00271)	
Post	-0.0168*** (0.00615)		
<i>Fixed Effects:</i>			
Year FE		✓	✓
Household FE			✓
Obs.	7,601,189	7,601,189	7,601,189
R ²	0.003	0.006	0.412
Mean(Dependent Var.)	0.662		

Renters increase their consumption after the LTV restriction

Do homeowners form a good control group for renters?

Possible issues

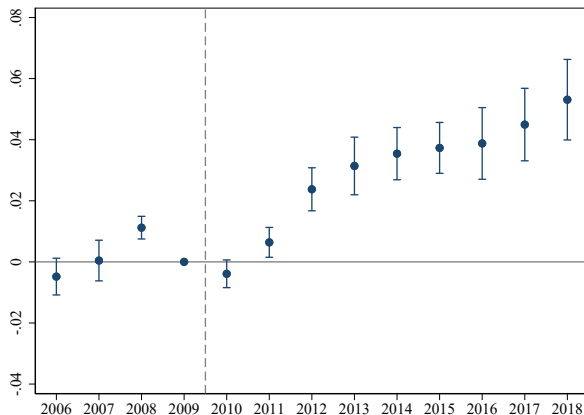
1. Homeowners are different from renters
→ **Parallel trends** before the restriction?

Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters

→ **Parallel trends** before the restriction?



Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters
 - Differences in characteristics start to affect consumption differentially in 2010

Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters

→ Differences in characteristics start to affect consumption differentially in 2010

	Consumption/Income					
	(1)	(2)	(3)	(4)	(5)	(6)
Renter $\times$ Post	0.0316*** (0.00451)	0.0293*** (0.00383)	0.0261*** (0.00374)	0.0412*** (0.00441)	0.0282*** (0.00361)	0.0385*** (0.00412)
<i>Fixed Effects:</i>						
Household FE	✓	✓	✓	✓	✓	✓
Age $\times$ Year FE	✓					
Deposit bins $\times$ Year FE		✓				
Financial Asset bins $\times$ Year FE			✓			
Total wealth bins $\times$ Year FE				✓		
Parental wealth bins $\times$ Year FE					✓	
Full Interacted FE						✓
Obs.	7,601,189	7,601,189	7,601,189	7,601,189	7,601,189	7,480,133
R ²	0.413	0.412	0.412	0.413	0.412	0.445
Mean(Dependent Var.)	0.660					

Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters
2. The restriction can limit homeowners' borrowing capacity
→ Suggesting that the consumption difference is due to homeowners' lower consumption

Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters
2. The restriction can limit homeowners' borrowing capacity
→ Suggesting that the consumption difference is due to homeowners' lower consumption

	Consumption/Income				
	(1) LTV<95%	(2) LTV<85%	(3) LTV<75%	(4) LTV<65%	(5)
Renter $\times$ Post	0.0258*** (0.00314)	0.0252*** (0.00302)	0.0242*** (0.00289)	0.0230*** (0.00274)	0.0308*** (0.00309)
<i>Fixed Effects:</i>					
Household FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	
Municipality $\times$ Year FE					✓
Obs.	7,230,463	7,062,467	6,809,292	6,444,576	7,601,189
R ²	0.411	0.411	0.413	0.415	0.417
Mean(Dependent Var.)	0.656				

Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters
2. The restriction can limit homeowners' borrowing capacity
3. The effect of homeownership on consumption changes over time
 - Housing wealth or house-related consumption
 - **Solution**: Within-renter comparison!

Do homeowners form a good control group for renters?

Possible issues

1. Homeowners are different from renters
2. The restriction can limit homeowners' borrowing capacity
3. The effect of homeownership on consumption changes over time
 - Housing wealth or house-related consumption
 - **Solution**: Within-renter comparison!

Within-renter comparison

- Are there renters who are less exposed to the restriction?
 - Renters want to remain as renters or are very similar to existing homeowners
 - How do we detect such renters with high ownership capacity?
 1. Split the sample into two: homeowners and renters
 2. Use XGBoost to predict the probability of being a homeowner by using 2006 information
- **Intuition**: **Renters with a high probability** have observables suggesting that they are similar to homeowners but still do not buy. Hence, they **are less affected**.

Within Renter Comparison

$$y_{it} = \beta (1 - \text{Homeowner Prob})_i \times \text{Post}_t + \gamma_1 (1 - \text{Homeowner Prob})_i + \gamma_2 \text{Post}_t + \epsilon_{it}$$

	Consumption/Income			
	(1)	(2)	(3)	(4)
(1-Homeowner prob.) $\times$ Post	0.0489*** (0.00650)	0.0320*** (0.00743)		
1(Homeowner prob.<80 _{pct}) $\times$ Post			0.0299*** (0.00523)	0.0173*** (0.00523)
<u>Fixed Effects:</u>				
Year FE	✓	✓	✓	✓
Household FE	✓	✓	✓	✓
Matching		✓		✓
Obs.	433,427	146,809	484,520	155,020
R ²	0.444	0.440	0.448	0.441
Mean(Dependent Var.)	0.606			

Results hold when making use of heterogeneity within renters

Why do renters increase their consumption?

- The aim of the policy is to decrease household leverage by increasing saving rates
→ Yet, we find the opposite effect! Why?

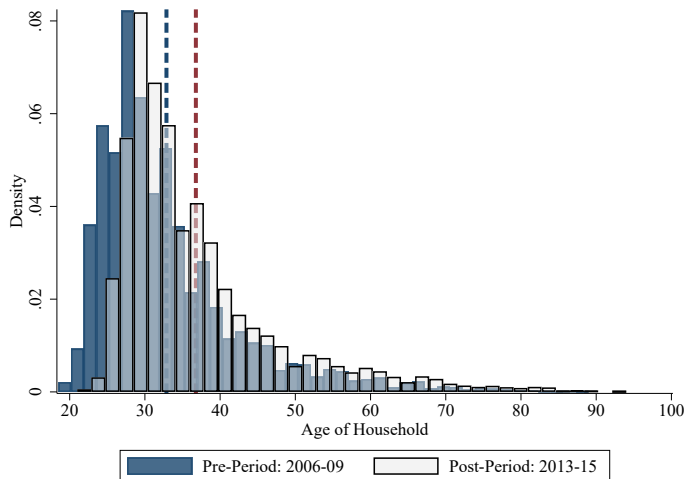
Why do renters increase their consumption?

- The aim of the policy is to decrease household leverage by increasing saving rates
→ Yet, we find the opposite effect! Why?
- **Renters may delay their home purchase and savings** (Karlman et al. (2024))
 - HHs need to reduce their consumption to accumulate savings
 - Leads to deviation in consumption smoothing, which is costly for renters
 - The disutility of this deviation can be larger than the benefits of an earlier home purchase
 - Low-income HHs that consider becoming homeowners delay both the home purchase and savings

Why do renters increase their consumption?

- The aim of the policy is to decrease household leverage by increasing saving rates
→ Yet, we find the opposite effect! Why?
- **Renters may delay their home purchase and savings** (Karlman et al. (2024))
 - HHs need to reduce their consumption to accumulate savings
 - Leads to deviation in consumption smoothing, which is costly for renters
 - The disutility of this deviation can be larger than the benefits of an earlier home purchase
 - Low-income HHs that consider becoming homeowners delay both the home purchase and savings
- **Two predictions**
 1. There must be a delay in home purchases
 2. The increase in consumption must be stronger for renters who expect a larger increase in their earnings

Age Distribution of Home Buying Cohorts



Homebuying cohorts get older

Earnings Potential Gap

Idea: How far a household is from the upper end of the income distribution among comparable households who are later in their career stages.

Earnings Potential Gap

Idea: How far a household is from the upper end of the income distribution among comparable households who are later in their career stages.

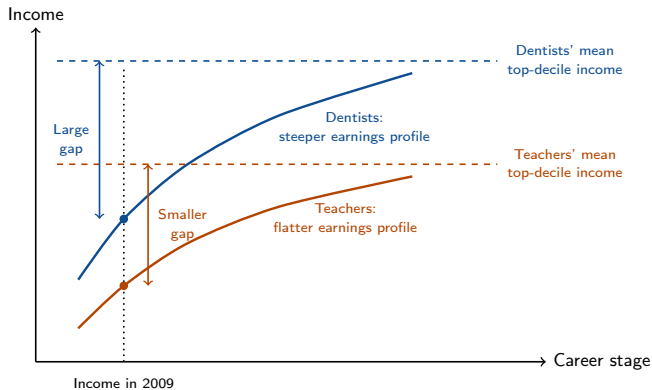
Comparable households:
education $\times$ labor-market
region $\times$ immigration background

$$\text{Gap}_i = \text{Top-decile income}_c - \text{Income}_{i,2009}$$

Earnings Potential Gap

Idea: How far a household is from the upper end of the income distribution among comparable households who are later in their career stages.

Comparable households:
education × labor-market
region × immigration background



$$\text{Gap}_i = \text{Top-decile income}_c - \text{Income}_{i,2009}$$

The gap is higher for HHs who are in the early stages of their careers and HHs whose cells have higher frontier income

Earnings Potential Gap

	Above top 10%	1 st Quart.	2 nd Quart.	3 rd Quart.	4 th Quart.
	(1)	(2)	(3)	(4)	(5)
Renter $\times$ Post	-0.00160 (0.00822)	0.00860*** (0.00297)	0.0169*** (0.00441)	0.0317*** (0.00494)	0.0401*** (0.00356)
<i>Fixed Effects:</i>					
Household FE	✓	✓	✓	✓	✓
Cell $\times$ Year FE	✓	✓	✓	✓	✓
Income bins $\times$ Year FE	✓	✓	✓	✓	✓
Obs.	544,254	1,690,597	1,866,771	1,822,874	1,668,886
R ²	0.424	0.407	0.434	0.465	0.470
Mean(Dependent Var.)	0.662				

Renters with greater anticipated income growth are more likely to delay the costly deviation from consumption smoothing

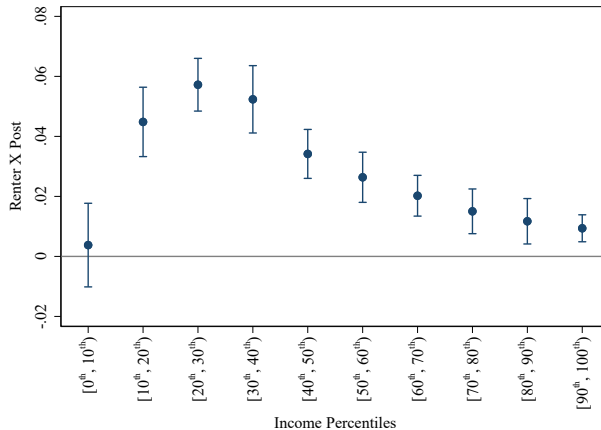
- The effect is stronger in regions with higher LTV ratios measured before the restriction
- The effect is insignificant for the lowest income decile. But stronger for the rest of low-income renters
- The renters increase their luxury spending, measured by expenditure elasticity
- No evidence for giving-up, as renters increase their effort in the labor market
- Renters move less and live in neighborhoods with higher income levels
- Renters increase their debt and reduce savings (deposits and financial assets)

- What about renters that purchased a home under the LTV restriction?
- We compare homebuyers in 2013 to homebuyers in 2009 and use homeowners as additional control for time effects
- Homebuyers in 2013 reduce their consumption significantly before the home purchase
- They also keep their consumption low after the purchase
 - They accumulate savings faster
 - No evidence for purchasing another home or locational sorting

- **Financial stability implications of macropru restrictions are not specific to homeowners**
- **LTV restrictions may stimulate short-term demand by increasing renters' consumption**

Thank You!

Income Heterogeneity - Levels



The effect is stronger for households with lower income

	MuniLTV Interaction	MuniLTV <25th perc.	MuniLTV 25th–75th perc.	MuniLTV >75th perc.
	(1)	(2)	(3)	(4)
Renter × Post × Municipality LTV	0.0036** (0.00177)			
Renter × Post	0.0306*** (0.00257)	0.0206*** (0.00422)	0.0289*** (0.00399)	0.0377*** (0.00314)
Post × Municipality LTV	0.0134*** (0.00214)			
<i>Fixed Effects:</i>				
HH FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Obs.	7,058,502	1,701,527	3,496,391	2,403,271
R ²	0.413	0.415	0.412	0.410
Mean(Municipality LTV)	0.923	0.810	0.921	1.030
Diff. to MuniLTV > 75th Coeff		0.0171***	0.0087*	
Diff. to MuniLTV < 25th Coeff			0.0084	

Renters that live in municipalities with higher LTV respond more strongly

Consumption Categories

	Cons. type/Income		Cons. type/Total Cons.	
	(1)	(2)	(3)	(4)
	$\epsilon < 1$	$\epsilon > 1$	$\epsilon < 1$	$\epsilon > 1$
Renter $\times$ Post	0.00324 (0.00252)	0.0199*** (0.00122)	-0.0192*** (0.00310)	0.0211*** (0.00363)
<i>Fixed Effects:</i>				
Household FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Obs.	7,601,189	7,601,189	7,601,189	7,601,189
R ²	0.579	0.444	0.547	0.542
Mean(Dependent Var.)	0.147	0.380	0.301	0.687

Substitution towards elastic consumption categories